Birzeit University
Department of Electrical & Computer Engineering
Second Semester, 2025/2026
ENCS5141 Intelligent Systems Lab
Assignment 1
Due Date May 15, 2026

Assignment Description

This assignment comprises two main components: **(1) Data Preprocessing and Feature Engineering**, and **(2) Comparative Evaluation of Classification Models** using the provided dataset.

Part 1: Data Preprocessing and Feature Engineering

Objectives

- Perform data exploration and preprocessing, including handling missing values, encoding categorical variables, cleaning noise, handling outliers, and scaling features where applicable.
- Apply feature selection and dimensionality reduction techniques.
- Handel class imbalance if exists.
- Assess the impact of preprocessing on classification performance.

Methodology

- **Exploration & Visualization:** Summarize dataset characteristics and visualize feature distributions and relationships.
- **Data Cleaning:** Address missing values and outliers using appropriate techniques.
- **Feature Engineering:** Encode categorical variables, normalize numerical features, and reduce dimensionality where applicable.
- **Evaluation:** Train a Random Forest model and compare performance on raw versus preprocessed data.

Expected Outcomes

- Quantitative comparison of model performance before and after preprocessing using standard metrics (e.g., accuracy, precision, recall).
- Analysis of the effect of preprocessing on model efficiency and predictive performance.

Part 2: Comparative Analysis of Classification Models

Objectives

- Compare the performance of **Random Forest (RF)**, **Support Vector Machine (SVM)**, and **Multilayer Perceptron (MLP)**.
- Examine the impact of hyperparameter tuning on model performance.

Methodology

- **Model Training:** Train RF, SVM, and MLP using the preprocessed dataset.
- **Evaluation:** Assess models using accuracy, precision, recall, and F1-score.
- **Analysis:** Compare models in terms of predictive performance and computational efficiency.
- **Parameter Study:** Investigate the effect of key hyperparameters.

Expected Outcomes

- Comparative analysis highlighting strengths and limitations of each model.
- Identification of the model achieving the best trade-off between accuracy and computational cost.

Deliverables

- Submit the implementation as a **.ipynb notebook**.
- Provide a structured report following the course guidelines (Attached).